

Here's a clean, practical project brief for practicing VLANs & Trunking without needing a Layer 3 switch:

Project Brief: VLANs and Trunking Lab (Router-on-a-Stick)

Company Name: Horizon Tech Solutions

You have been hired to segment the company network using **VLANs** for better security and performance.

Requirements:

The company has 3 departments:

VLAN ID	Department	Number of PCs	Subnet
10	Management	4 PCs	192.168.10.0/24
20	Sales	6 PCs	192.168.20.0/24
30	IT / Engineering	5 PCs	192.168.30.0/24

Devices:

- 1 x Router (2911 or 4331)
- 2 x Layer 2 Switches (2960)
- 15 x PCs (distributed across departments)

Topology Requirements:

- Connect both switches to the router using **trunk links**.
 - Use **one physical interface** on the router for all VLANs (Router-on-a-Stick configuration).
 - PCs in different VLANs should **not** be able to communicate with each other unless going through the router.
 - Management PCs should be able to ping Sales and IT PCs (Inter-VLAN routing).
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Tasks You Must Complete:

1. Create the 3 VLANs on both switches:
 - VLAN 10 - Management
 - VLAN 20 - Sales
 - VLAN 30 - IT
 2. Assign switch ports to the correct VLANs (Access Ports).
 3. Configure **Trunk ports** between:
 - Switch1 ↔ Switch2
 - Switch1 ↔ Router
 - Switch2 ↔ Router (if needed)
 4. Configure **Router-on-a-Stick** on the router:
 - Create subinterfaces (e.g. GigabitEthernet0/0.10 , GigabitEthernet0/0.20 , GigabitEthernet0/0.30)
 - Assign correct IP addresses and VLAN encapsulation
 5. Assign correct IP addresses and default gateways to all PCs.
 6. Test:
 - PCs in the same VLAN can ping each other
 - PCs in different VLANs can ping each other (via router)
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Would you like me to also give you:

- A detailed **IP addressing table**?
- Step-by-step configuration commands?
- Or a diagram description?

Just say what you need and I'll provide it. This is an excellent lab for your GitHub repo!